

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech (IT) Examination

March - April 2018

IT210 Data Communication

Date: 06.04.2018, Friday

Time: 01.30 p.m. To 04.30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Answer the following questions. [07]

- a. Define the term: Protocol. [01]
- b. Which of the following topology requires multipoint connection? [01]
1) Mesh 2) Star 3) Ring 4) Bus
- c. List out the protocols used in the Network (Internet) layer of TCP/IP model. [01]
- d. Define the terms: Frequency and Bandwidth. [02]
- e. Differentiate between SDM and TDM. [02]

Q-2.a What is the difference between Physical address and Logical address of node in network? [02]

Q-2.b Draw a hybrid topology with a star backbone with three bus networks. [04]

Q-2.c Answer any TWO questions. [08]

- (i) Explain the simplex, half-duplex and full-duplex communication in detail.
- (ii) Draw TCP/IP Model of Network and Explain the different layers of TCP/IP in detail.
- (iii) List and Explain different types of topology in detail.

Q-3 Answer any TWO questions. [14]

- a. Draw an OSI Model of Network and Explain the different layers of OSI in detail.
- b. Consider the composite periodic signal with frequencies 3Hz, 9Hz and 27Hz with amplitude 5V, 3V and 1V consecutively. Show the decomposition of same signal in the time and frequency domain.
- c. Explain Local Area Network, Metropolitan Area Network and Wide Area Network in detail.

SECTION – II**Q - 4 Answer the following questions. [07]**

- a. Define the term: wavelength. [01]
- b. Which of following is/are digital modulations scheme? [01]
 1) ASK 2) FSK 3) PSK 4) All of these
- c. The principle of optical fiber communication is [01]
 1) Total Internal Reflection 2) Radiation
 3) Refraction 4) None of these
- d. Which of the following blocks are present in Pulse code modulation? [01]
 1) Sampler 2) Quantizer 3) Encoder 4) All of these
- e. Electronics Industry Association provides RS-232 interface standards. **True/False** [01]
- f. Which form of multiplexing is most complicated technically? Which form is least complicated? [02]

Q-5 a. Answer any THREE questions. [12]

- (i) Explain the Time division multiplexing process in detail with suitable diagram.
- (ii) List out limitations of RS-232 interface. Draw well labeled diagram of functioning of EIA-232 in synchronous full duplex mode.
- (iii) Write a short note on null modem. Draw suitable diagram.
- (iv) Compare Twisted pair cable, coaxial cable and fiber optic cable.

Q-5 b. What is the difference between a DTE and a DCE? [02]**Q-6 Answer any TWO questions. [14]**

- a. Compare various analog modulation (AM, FM, PM) techniques.
- b. Explain Electrical & Mechanical characteristics of DTE-DCE interface.
- c. Explain Amplitude shift keying and Frequency shift keying in detail.
